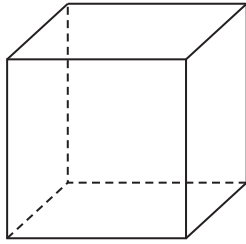


8 The diagram shows a solid cube placed on a horizontal table.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The pressure on the table due to the cube is  $3.5 \text{ newtons/cm}^2$

The force exerted by the cube on the table is 504 newtons.

Show that the total surface area of the cube is less than  $900 \text{ cm}^2$